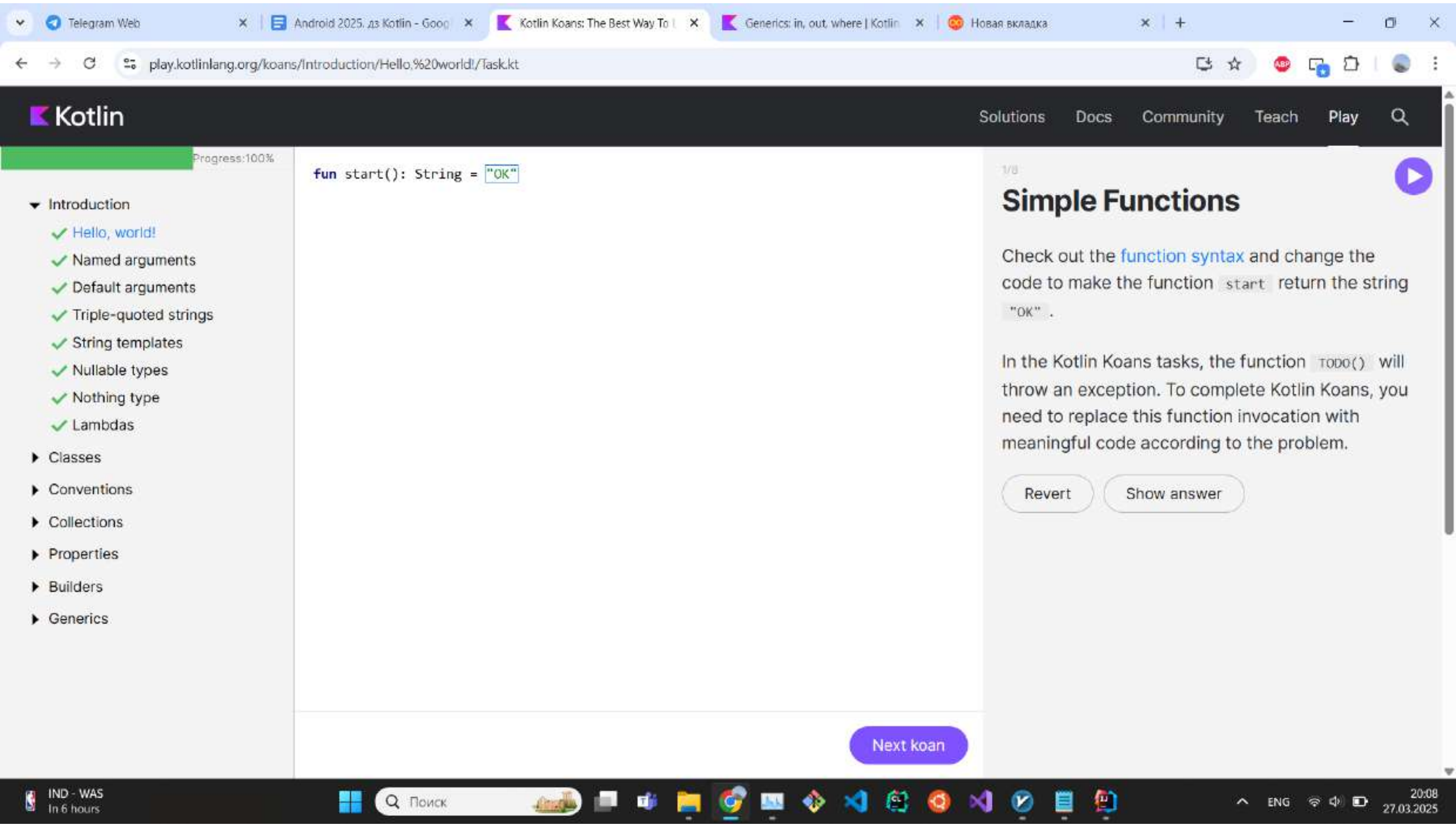
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fun start(): String = "OK"
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Simple Functions
Check out the function syntax and change the code to make the function start return the string "OK".
In the Kotlin Koans tasks, the function TODO() will throw an exception. To complete Kotlin Koans, you need to replace this function invocation with meaningful code according to the problem.
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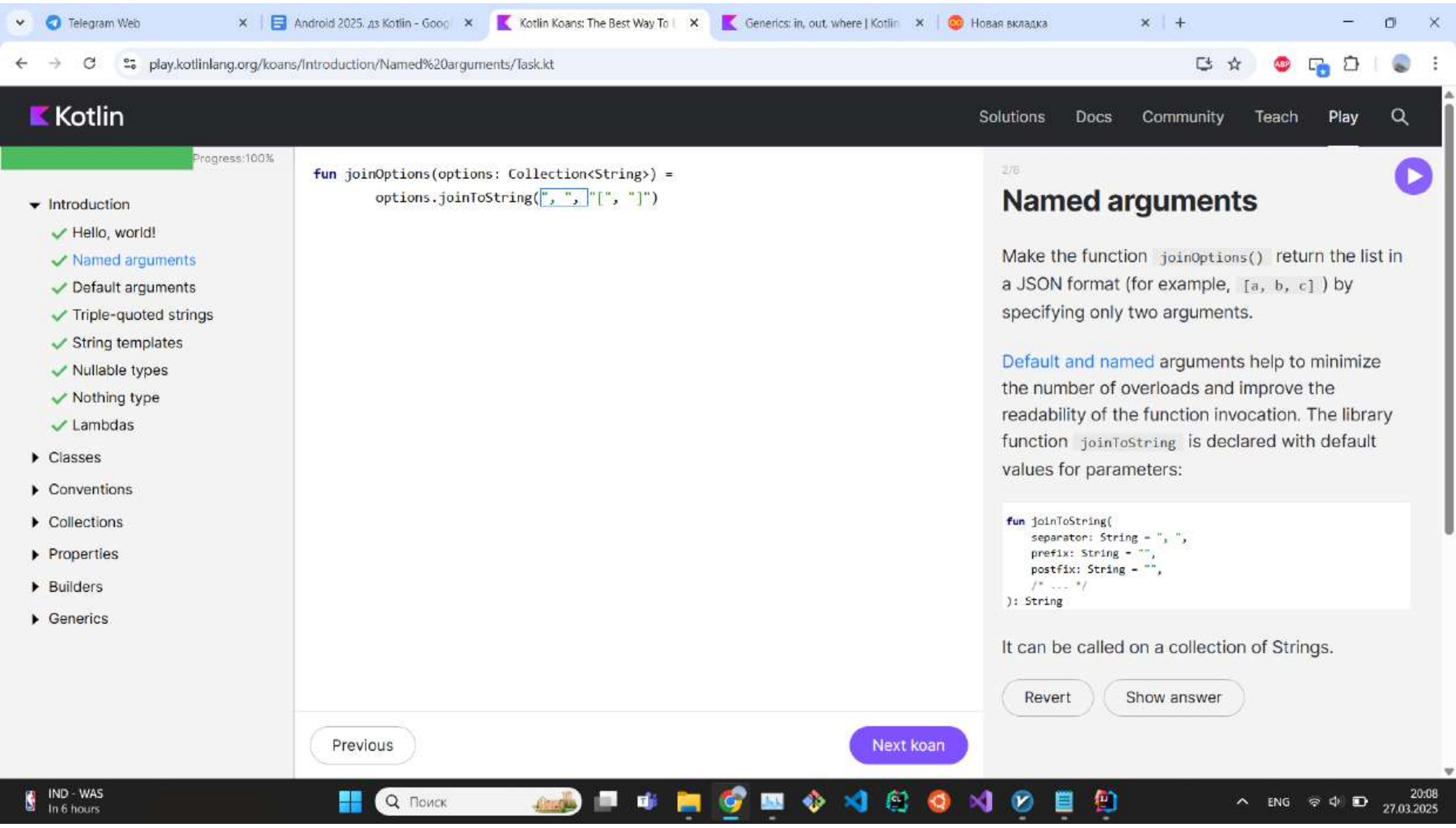
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fun joinOptions(options: Collection<String>) =
options.joinToString(", ", "[", "]")
2/8
Named arguments
Make the function joinOptions() return the list in a JSON format (for example, [a, b, c]) by specifying only two arguments.
Default and named arguments help to minimize the number of overloads and improve the readability of the function invocation. The library function joinToString is declared with default values for parameters:
fun joinToString(
separator: String = ", ",
prefix: String = "",
postfix: String = "",
/* ... */
): String
It can be called on a collection of Strings.
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```kotlin
fun foo(name: String, number: Int = 42, toUpperCase: Boolean = false) =
        (if (toUpperCase) name.uppercase() else name) + number

fun useFoo() = listOf(
        foo("a"),
        foo("b", number = 1),
        foo("c", toUpperCase = true),
        foo(name = "d", number = 2, toUpperCase = true)
)
```

3/8

Default arguments

Imagine you have several overloads of 'foo()' in Java:

```java
public String foo(String name, int number, boolean toUpperCase) {
    return (toUpperCase ? name.toUpperCase() : name) + number;
}
public String foo(String name, int number) {
    return foo(name, number, false);
}
public String foo(String name, boolean toUpperCase) {
    return foo(name, 42, toUpperCase);
}
public String foo(String name) {
    return foo(name, 42);
}
```

You can replace all these Java overloads with one function in Kotlin. Change the declaration of the `foo` function in a way that makes the code using `foo` compile. Use default and named arguments.

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```kotlin
const val question = "life, the universe, and everything"
const val answer = 42

val tripleQuotedString = """
    #question = "$question"
    #answer = $answer""".trimMargin("#")

fun main() {
    println(tripleQuotedString)
}
```

4/8

Triple-quoted strings

Learn about the different string literals and string templates in Kotlin.

You can use the handy library functions `trimIndent` and `trimMargin` to format multiline triple-quoted strings in accordance with the surrounding code.

Replace the `trimIndent` call with the `trimMargin` call taking `#` as the prefix value so that the resulting string doesn't contain the prefix character.

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```kotlin
val month = "(JAN|FEB|MAR|APR|MAY|JUN|JUL|AUG|SEP|OCT|NOV|DEC)"

fun getPattern(): String = """\d{2} """ + month + """ \d{4}"""
```

5/8

String templates

Triple-quoted strings are not only useful for multiline strings but also for creating regex patterns as you don't need to escape a backslash with a backslash.

The following pattern matches a date in the format `13.06.1992` (two digits, a dot, two digits, a dot, four digits):

```kotlin
fun getPattern() = """\d{2}\.\d{2}\.\d{4}"""
```

Using the `month` variable, rewrite this pattern in such a way that it matches the date in the format `13 JUN 1992` (two digits, one whitespace, a month abbreviation, one whitespace, four digits).

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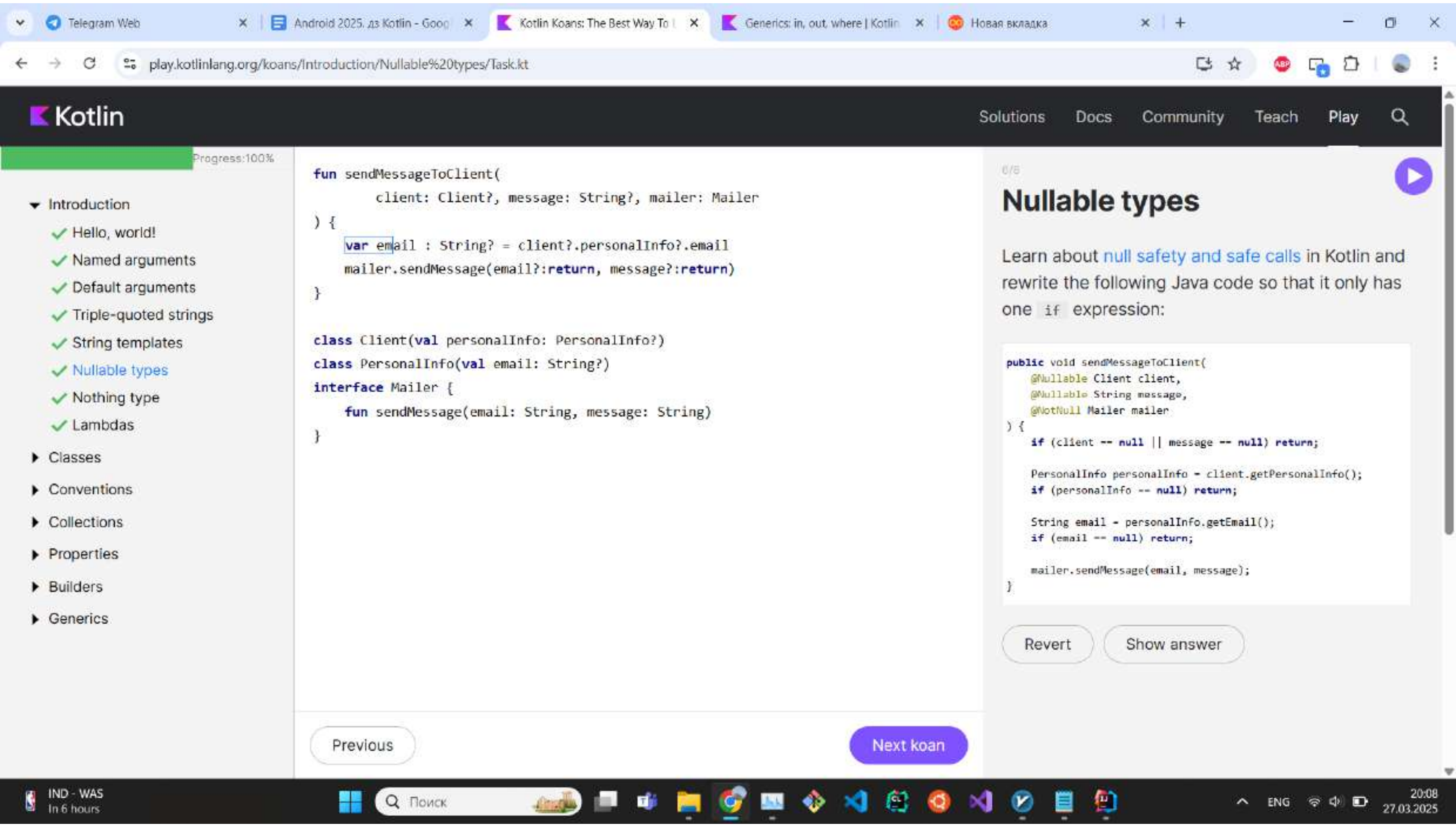
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fun sendMessageToClient(
client: Client?, message: String?, mailer: Mailer
) {
var email : String? = client?.personalInfo?.email
mailer.sendMessage(email?:return, message?:return)
}
class Client(val personalInfo: PersonalInfo?)
class PersonalInfo(val email: String?)
interface Mailer {
fun sendMessage(email: String, message: String)
}
Nullable types
Learn about null safety and safe calls in Kotlin and rewrite the following Java code so that it only has one if expression:
public void sendMessageToClient(
@Nullable Client client,
@Nullable String message,
@NotNull Mailer mailer
) {
if (client == null || message == null) return;
PersonalInfo personalInfo = client.getPersonalInfo();
if (personalInfo == null) return;
String email = personalInfo.getEmail();
if (email == null) return;
mailer.sendMessage(email, message);
}
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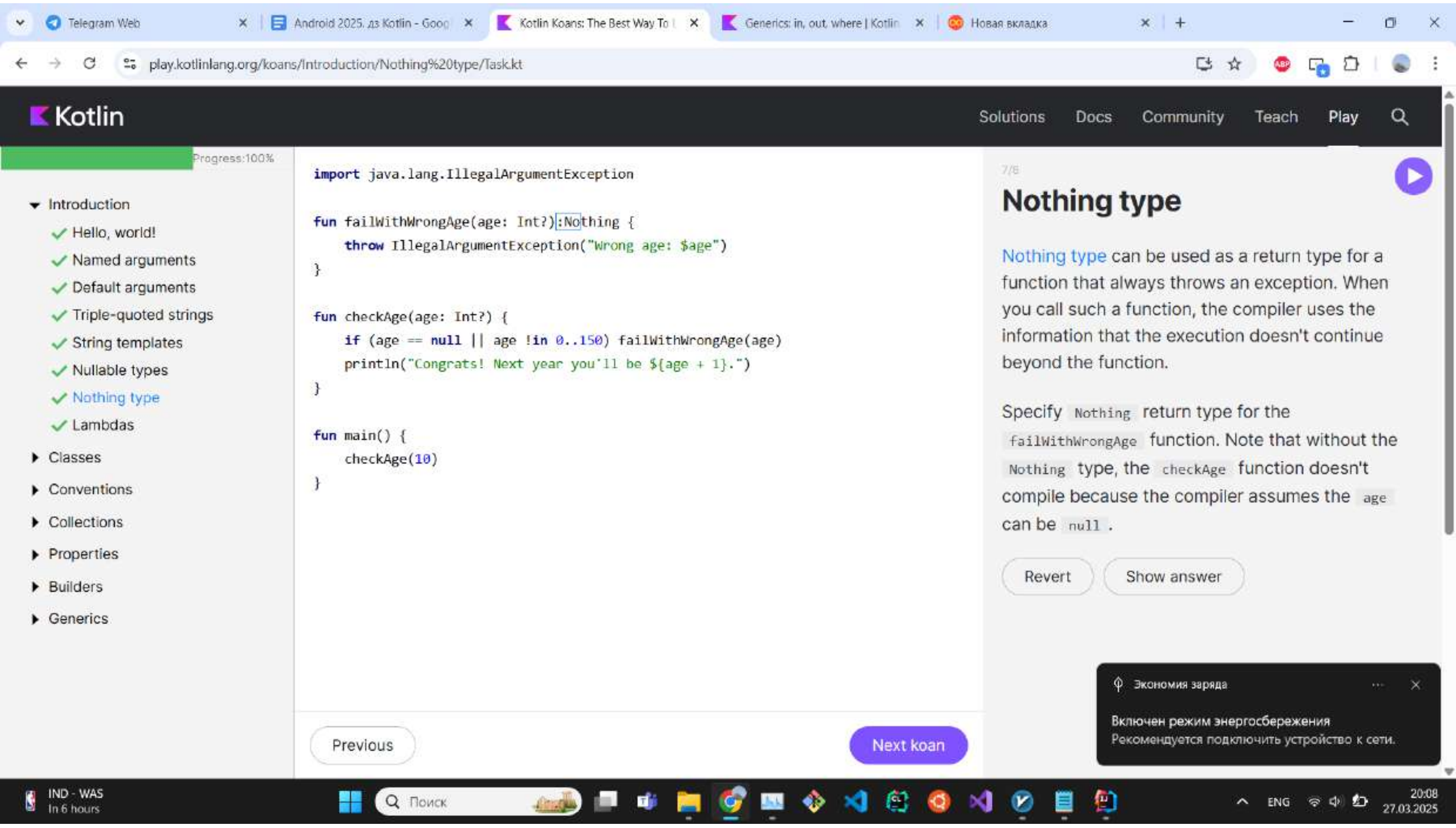

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import java.lang.IllegalArgumentException
fun failWithWrongAge(age: Int?):Nothing {
throw IllegalArgumentException("Wrong age: $age")
}
fun checkAge(age: Int?) {
if (age == null || age !in 0..150) failWithWrongAge(age)
println("Congrats! Next year you'll be ${age + 1}.")
}
fun main() {
checkAge(10)
}
Nothing type
Nothing type can be used as a return type for a function that always throws an exception. When you call such a function, the compiler uses the information that the execution doesn't continue beyond the function.
Specify Nothing return type for the failWithWrongAge function. Note that without the Nothing type, the checkAge function doesn't compile because the compiler assumes the age can be null.
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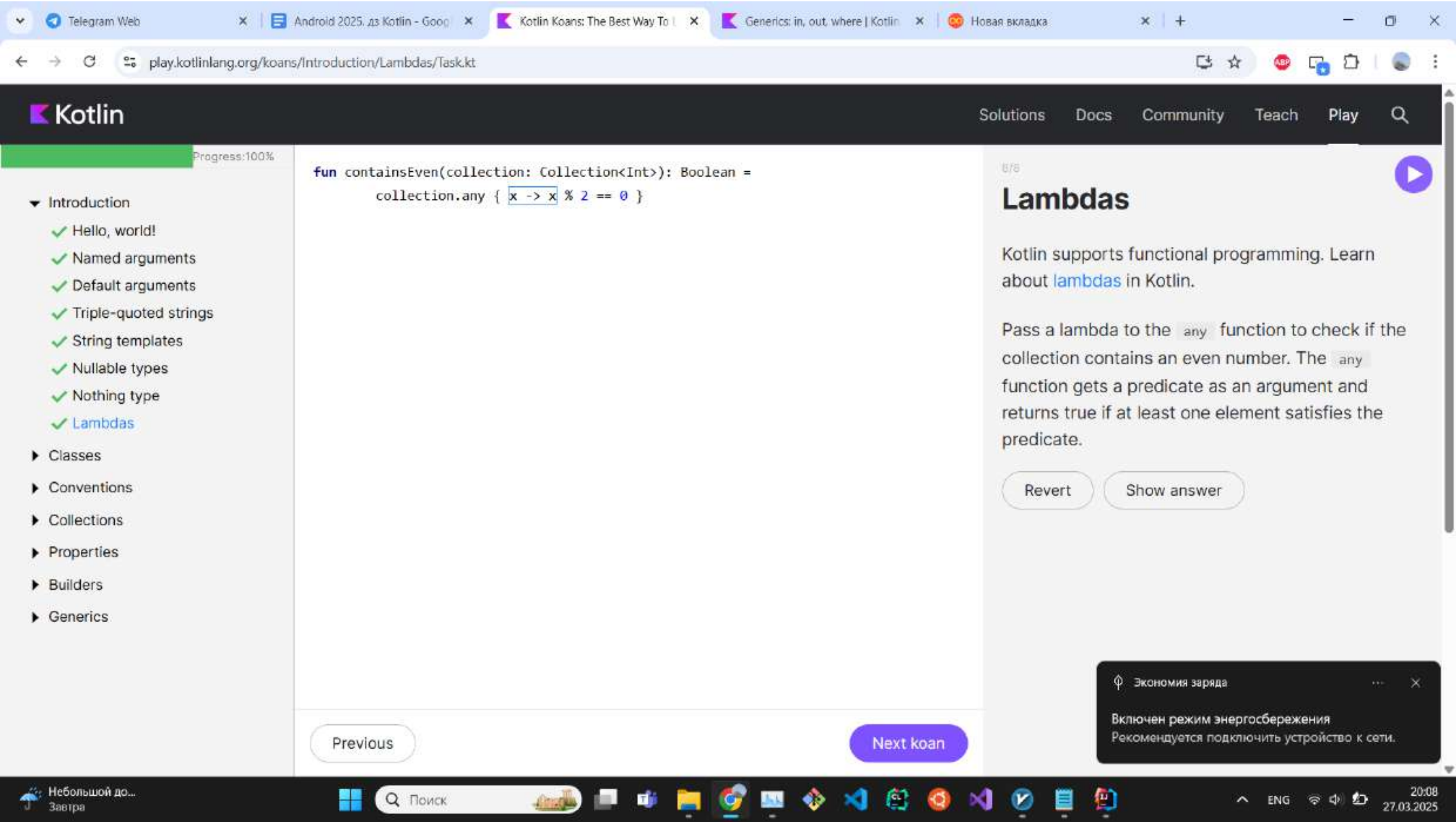
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fun containsEven(collection: Collection<Int>): Boolean =
collection.any { x -> x % 2 == 0 }
Lambdas
Kotlin supports functional programming. Learn about lambdas in Kotlin.
Pass a lambda to the any function to check if the collection contains an even number. The any function gets a predicate as an argument and returns true if at least one element satisfies the predicate.
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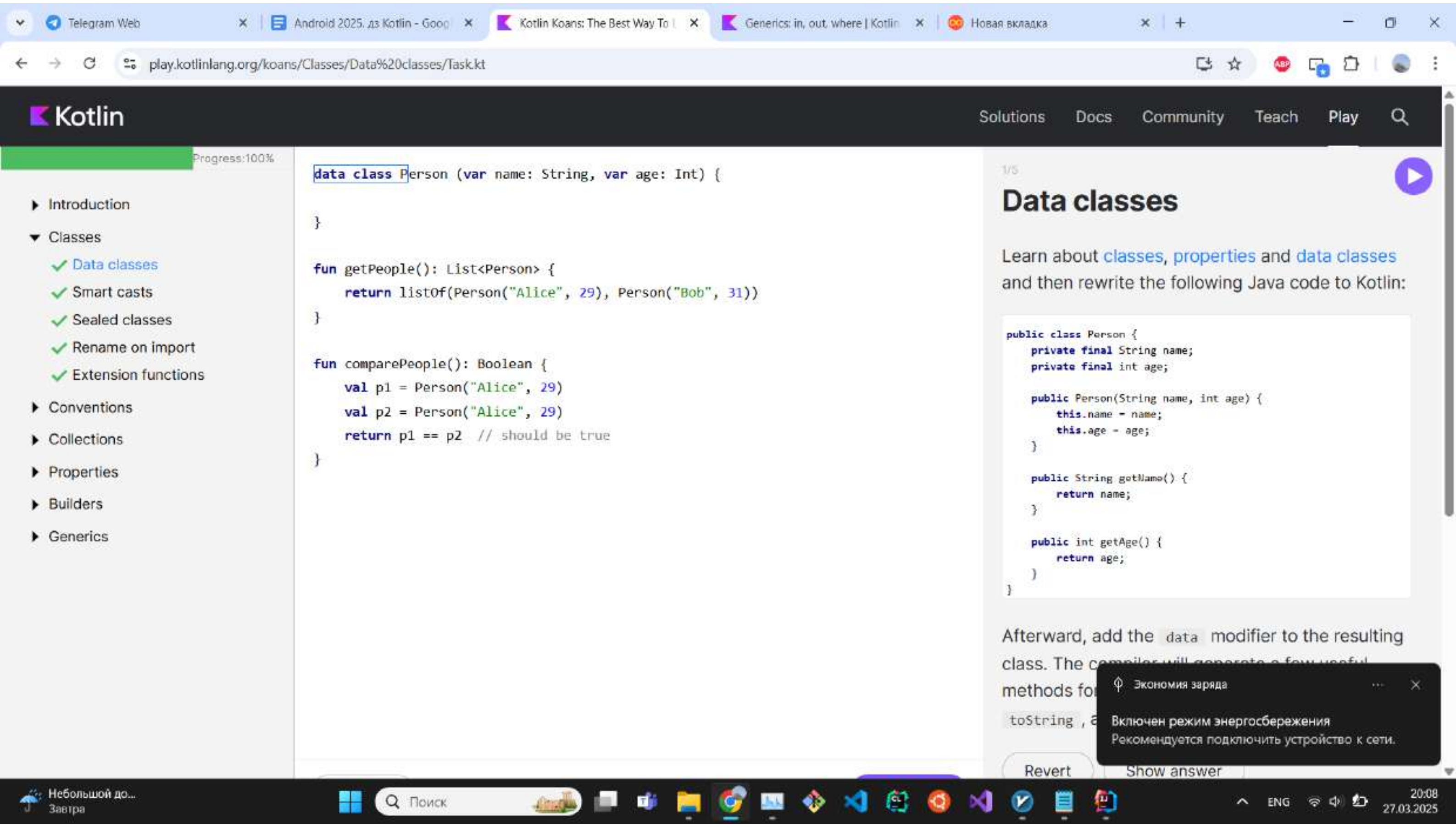
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data class Person (var name: String, var age: Int) {
}
fun getPeople(): List<Person> {
return listOf(Person("Alice", 29), Person("Bob", 31))
}
fun comparePeople(): Boolean {
val p1 = Person("Alice", 29)
val p2 = Person("Alice", 29)
return p1 == p2 // should be true
}
1/5
Data classes
Learn about classes, properties and data classes and then rewrite the following Java code to Kotlin:
public class Person {
private final String name;
private final int age;
public Person(String name, int age) {
this.name = name;
this.age = age;
}
public String getName() {
return name;
}
public int getAge() {
return age;
}
}
Afterward, add the data modifier to the resulting class. The c
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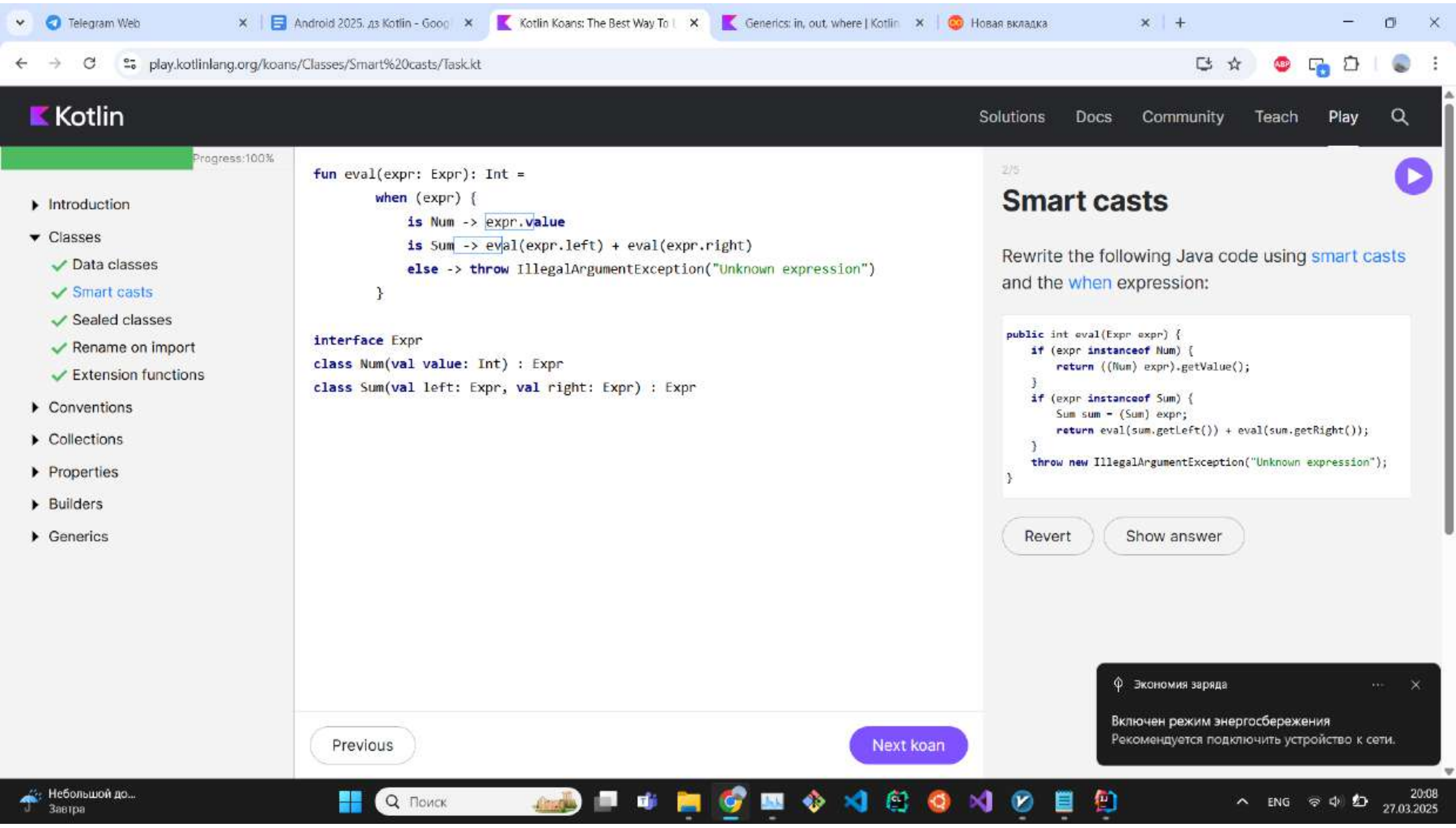
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fun eval(expr: Expr): Int =
when (expr) {
is Num -> expr.value
is Sum -> eval(expr.left) + eval(expr.right)
else -> throw IllegalArgumentException("Unknown expression")
}
interface Expr
class Num(val value: Int) : Expr
class Sum(val left: Expr, val right: Expr) : Expr
2/5
Smart casts
Rewrite the following Java code using smart casts and the when expression:
public int eval(Expr expr) {
if (expr instanceof Num) {
return ((Num) expr).getValue();
}
if (expr instanceof Sum) {
Sum sum = (Sum) expr;
return eval(sum.getLeft()) + eval(sum.getRight());
}
throw new IllegalArgumentException("Unknown expression");
}
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```kotlin
fun eval(expr: Expr): Int =
        when (expr) {
            is Num -> expr.value
            is Sum -> eval(expr.left) + eval(expr.right)
            else -> throw IllegalArgumentException("Unknown expression")
        }

sealed interface Expr
class Num(val value: Int) : Expr
class Sum(val left: Expr, val right: Expr) : Expr
```

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Sealed classes

Reuse your solution from the previous task, but replace the interface with the `sealed` interface. Then you no longer need the `else` branch in `when`.

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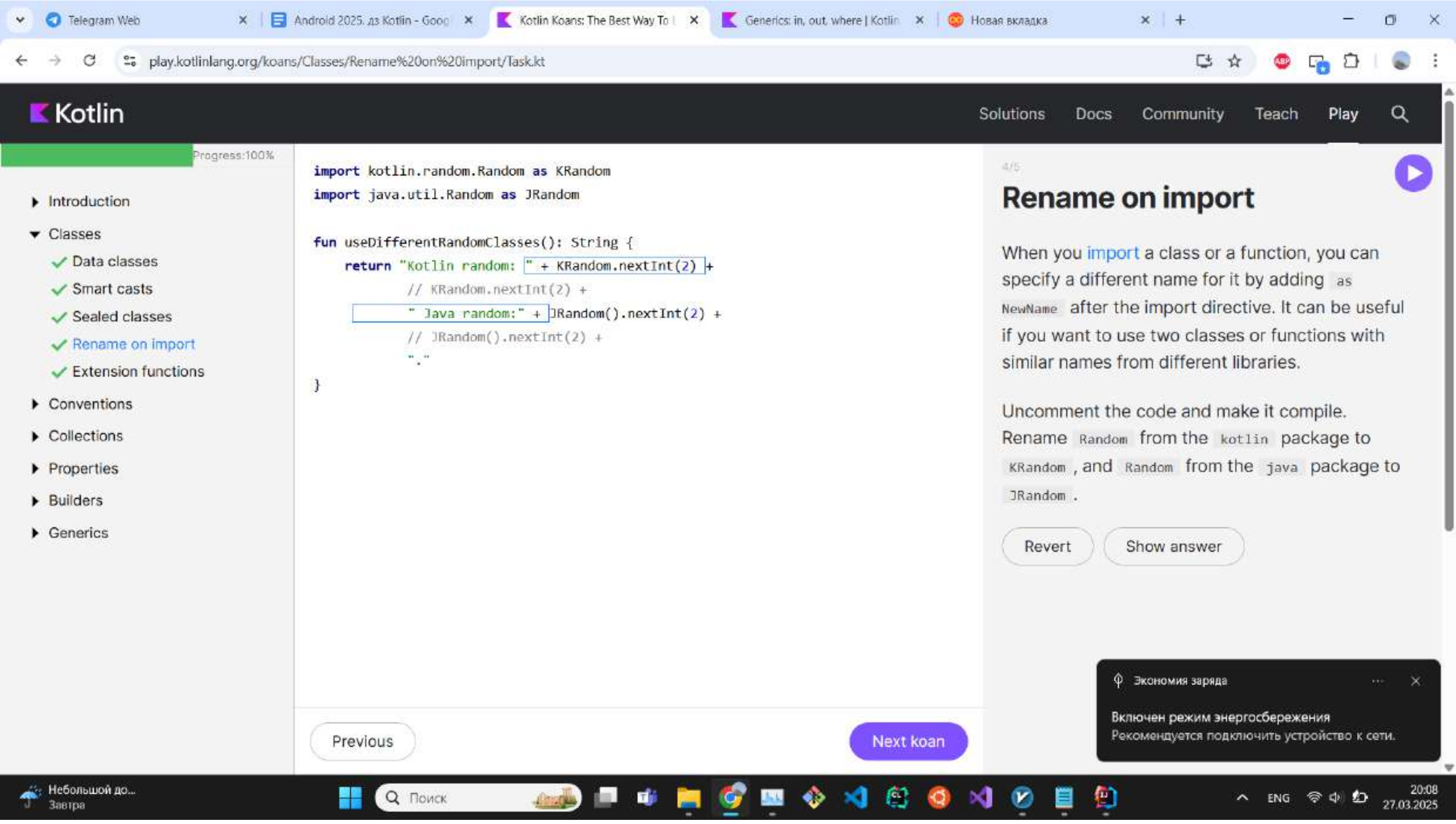

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import kotlin.random.Random as KRandom
import java.util.Random as JRandom
fun useDifferentRandomClasses(): String {
return "Kotlin random: " + KRandom.nextInt(2) +
// KRandom.nextInt(2) +
" Java random:" + JRandom().nextInt(2) +
// JRandom().nextInt(2) +
"."
}
4/5
Rename on import
When you import a class or a function, you can specify a different name for it by adding as NewName after the import directive. It can be useful if you want to use two classes or functions with similar names from different libraries.
Uncomment the code and make it compile. Rename Random from the kotlin package to KRandom, and Random from the java package to JRandom.
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```
fun Int.r(): RationalNumber = RationalNumber(this, 1)

fun Pair<Int, Int>.r(): RationalNumber = RationalNumber(this.first, this.second)

data class RationalNumber(val numerator: Int, val denominator: Int)
```

5/5

Extension functions

Learn about extension functions. Then implement the extension functions `Int.r()` and `Pair.r()` and make them convert `Int` and `Pair` to a `RationalNumber`.

`Pair` is a class defined in the standard library:

```
data class Pair<out A, out B>(
    val first: A,
    val second: B
)
```

In the case of `Int`, the denominator is 1.

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```kotlin
data class MyDate(val year: Int, val month: Int, val dayOfMonth: Int) : Comparable<MyD
{
    override fun compareTo(date2: MyDate): Int {
        if (this.year != date2.year)
            return year.compareTo(date2.year)
        if (this.month != date2.month)
            return month.compareTo(date2.month)
        return this.dayOfMonth.compareTo(date2.dayOfMonth)
    }
}

fun test(date1: MyDate, date2: MyDate) {
    // this code should compile:
    println(date1 < date2)
}
```

1/5

Comparison

Learn about operator overloading and how the different conventions for operations like `==`, `<`, `+` work in Kotlin. Add the function `compareTo` to the class `MyDate` to make it comparable. After this, the code below `date1 < date2` should start to compile.

Note that when you override a member in Kotlin, the `override` modifier is mandatory.

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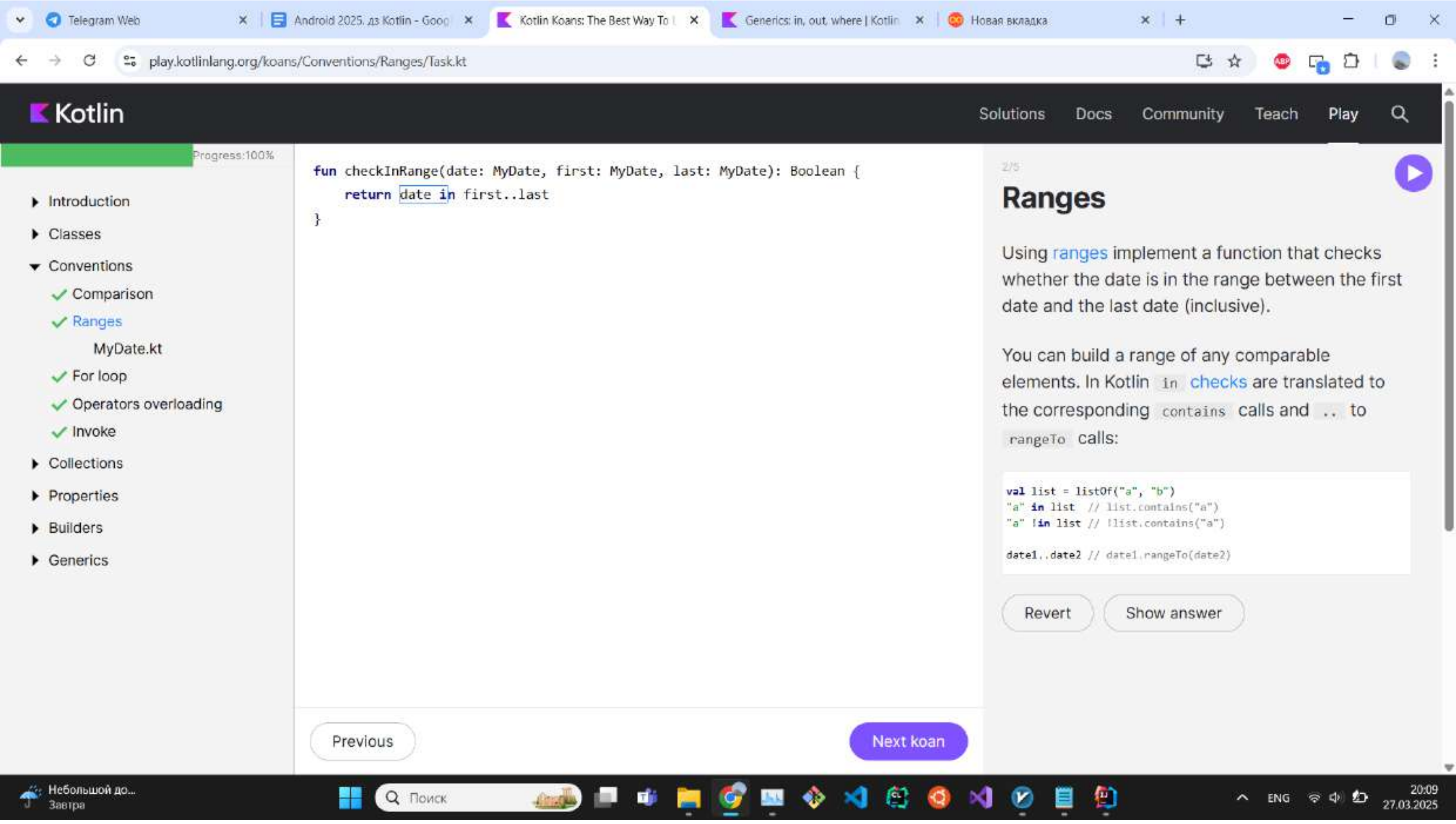
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fun checkInRange(date: MyDate, first: MyDate, last: MyDate): Boolean {
return date in first..last
}
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Ranges
Using ranges implement a function that checks whether the date is in the range between the first date and the last date (inclusive).
You can build a range of any comparable elements. In Kotlin in checks are translated to the corresponding contains calls and .. to rangeTo calls:
val list = listOf("a", "b")
"a" in list // list.contains("a")
"a" !in list // !list.contains("a")
date1..date2 // date1.rangeTo(date2)
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```kotlin
class DateRange(val start: MyDate, val end: MyDate) : Iterable<MyDate> {
    override fun iterator(): Iterator<MyDate> {
        return object : Iterator<MyDate> {
            var date = start
            override operator fun hasNext(): Boolean {
                return date <= end
            }
            override operator fun next(): MyDate {
                val s = date
                date = date.followingDate()
                return s
            }
        }
    }
}

fun iterateOverDateRange(firstDate: MyDate, secondDate: MyDate, handler: (MyDate) -> U
    for (date in firstDate..secondDate) {
        handler(date)
    }
}
```

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For loop

A Kotlin for loop can iterate through any object if the corresponding `iterator` member or extension function is available.

Make the class `DateRange` implement `Iterable<MyDate>`, so that it can be iterated over. Use the function `MyDate.followingDate()` defined in `DateUtil.kt`; you don't have to implement the logic for finding the following date on your own.

Use an object expression which plays the same role in Kotlin as an anonymous class in Java.

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```kotlin
import TimeInterval.*

data class MyDate(val year: Int, val month: Int, val dayOfMonth: Int)

// Supported intervals that might be added to dates:
enum class TimeInterval { DAY, WEEK, YEAR }

operator fun MyDate.plus(timeInterval: TimeInterval): MyDate =
    this.addTimeIntervals(timeInterval, 1)

operator fun TimeInterval.times(num: Int): Pair<TimeInterval, Int> = Pair(this, num)

operator fun MyDate.plus(interval: Pair<TimeInterval, Int>): MyDate =
    this.addTimeIntervals(interval.first, interval.second)


fun task1(today: MyDate): MyDate {
    return today + YEAR + WEEK
}

fun task2(today: MyDate): MyDate {
    return today + YEAR * 2 + WEEK * 3 + DAY * 5
}
```

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Operators overloading

Implement date arithmetic and support adding years, weeks, and days to a date. You could write the code like this: `date + YEAR * 2 + WEEK * 3 + DAY * 15`.

First, add the extension function `plus()` to `MyDate`, taking the `TimeInterval` as an argument. Use the utility function `MyDate.addTimeIntervals()` declared in `DateUtil.kt`

Then, try to support adding several time intervals to a date. You may need an extra class.

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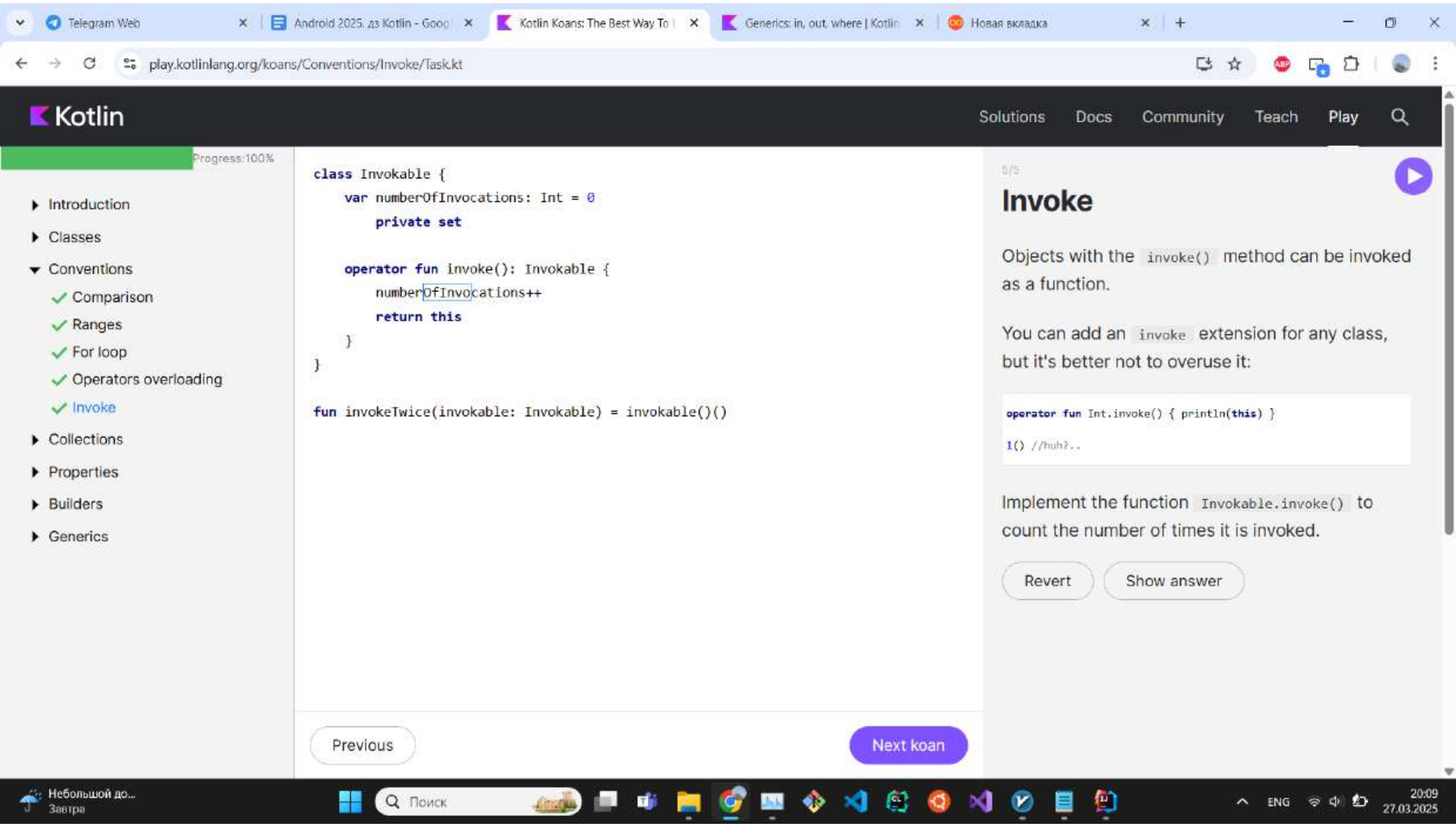
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class Invokable {
var numberOfInvocations: Int = 0
private set
operator fun invoke(): Invokable {
numberOfInvocations++
return this
}
}
fun invokeTwice(invokable: Invokable) = invokable()()
5/5
Invoke
Objects with the invoke() method can be invoked as a function.
You can add an invoke extension for any class, but it's better not to overuse it:
operator fun Int.invoke() { println(this) }
1() //huh?..
Implement the function Invokable.invoke() to count the number of times it is invoked.
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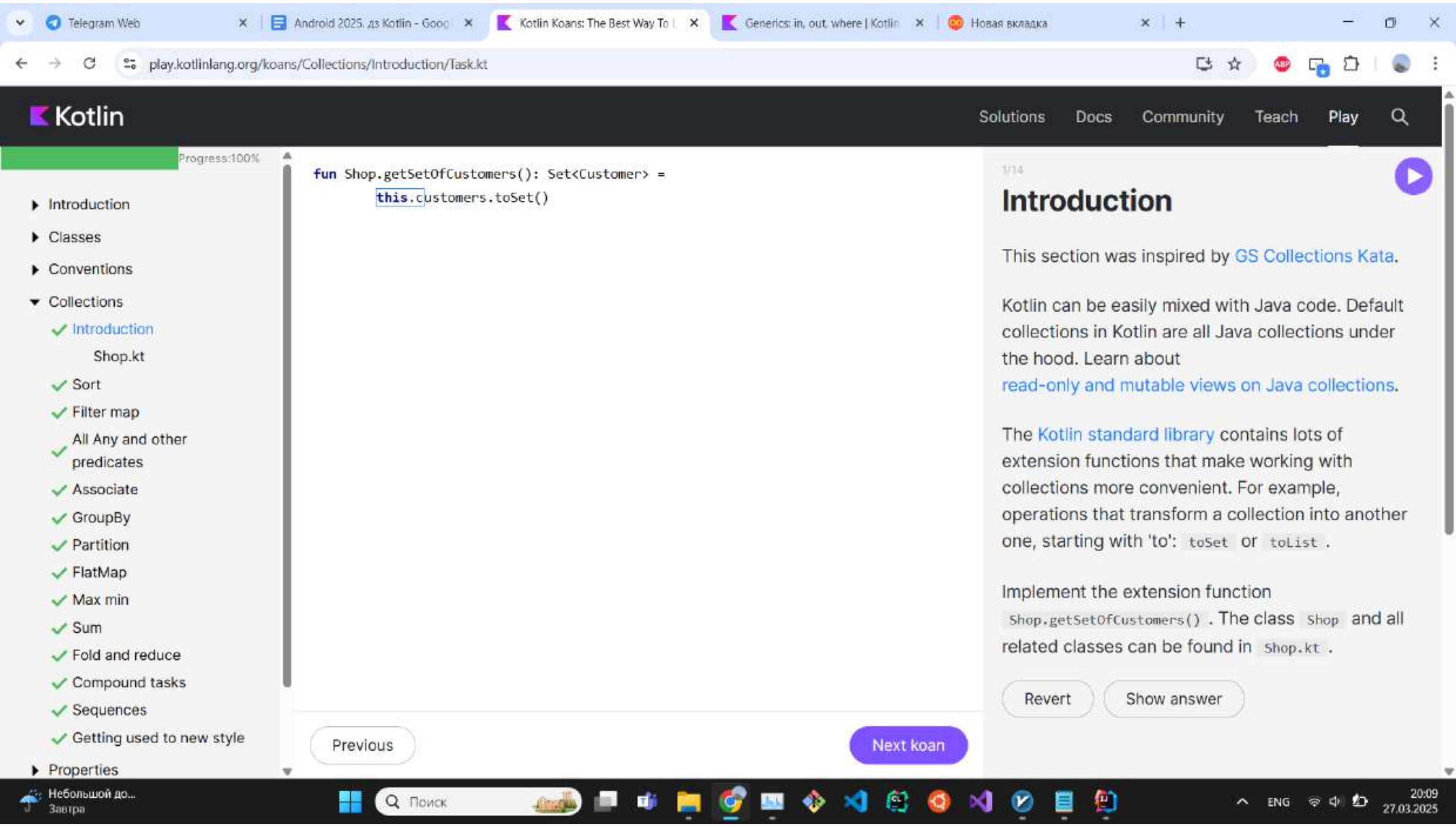

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fun Shop.getSetOfCustomers(): Set<Customer> =
this.customers.toSet()
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Introduction
This section was inspired by GS Collections Kata.
Kotlin can be easily mixed with Java code. Default collections in Kotlin are all Java collections under the hood. Learn about read-only and mutable views on Java collections.
The Kotlin standard library contains lots of extension functions that make working with collections more convenient. For example, operations that transform a collection into another one, starting with 'to': toSet or toList.
Implement the extension function Shop.getSetOfCustomers(). The class Shop and all related classes can be found in Shop.kt.
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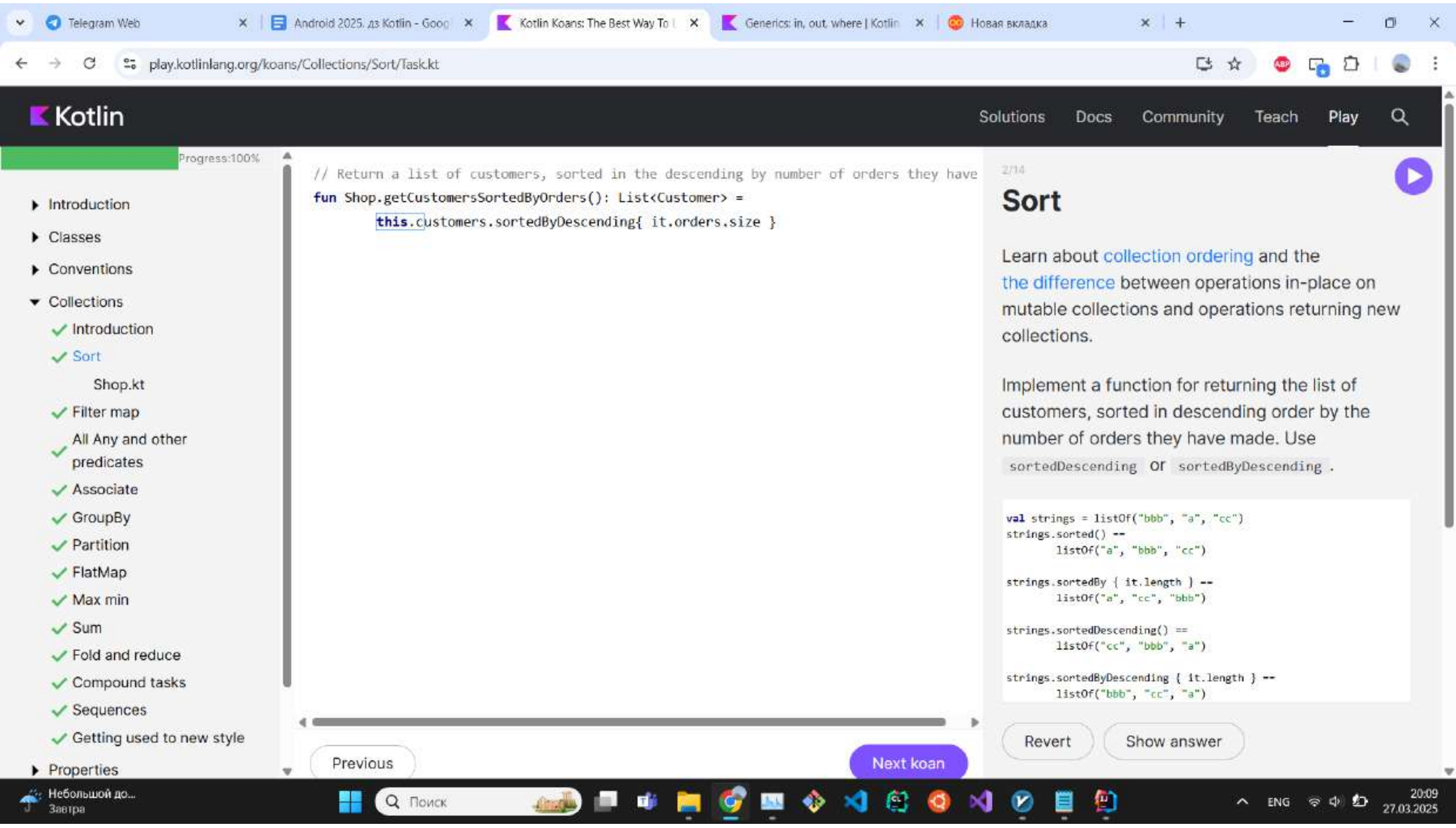
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// Return a list of customers, sorted in the descending by number of orders they have
fun Shop.getCustomersSortedByOrders(): List<Customer> =
this.customers.sortedByDescending{ it.orders.size }
2/14
Sort
Learn about collection ordering and the the difference between operations in-place on mutable collections and operations returning new collections.
Implement a function for returning the list of customers, sorted in descending order by the number of orders they have made. Use sortedDescending or sortedByDescending.
val strings = listOf("bbb", "a", "cc")
strings.sorted() ==
listOf("a", "bbb", "cc")
strings.sortedBy { it.length } ==
listOf("a", "cc", "bbb")
strings.sortedDescending() ==
listOf("cc", "bbb", "a")
strings.sortedByDescending { it.length } ==
listOf("bbb", "cc", "a")
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```kotlin
// Find all the different cities the customers are from
fun Shop.getCustomerCities(): Set<City> =
        this.customers.map{it.city}.toSet()

// Find the customers living in a given city
fun Shop.getCustomersFrom(city: City): List<Customer> =
        this.customers.filter{it.city == city}
```

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3/14

Filter; map

Learn about mapping and filtering a collection.

Implement the following extension functions using the `map` and `filter` functions:

- Find all the different cities the customers are from
- Find the customers living in a given city

```kotlin
val numbers = listOf(1, -1, 2)
numbers.filter { it > 0 } == listOf(1, 2)
numbers.map { it * it } == listOf(1, 1, 4)
```

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```kotlin
// Return true if all customers are from a given city
fun Shop.checkAllCustomersAreFrom(city: City): Boolean =
        this.customers.all{it.city == city}

// Return true if there is at least one customer from a given city
fun Shop.hasCustomerFrom(city: City): Boolean =
        this.customers.any{it.city == city}

// Return the number of customers from a given city
fun Shop.countCustomersFrom(city: City): Int =
        this.customers.count{it.city == city}

// Return a customer who lives in a given city, or null if there is none
fun Shop.findCustomerFrom(city: City): Customer? =
        this.customers.find{it.city == city}
```

4/14

All, Any, and other predicates

Learn about testing predicates and retrieving elements by condition.

Implement the following functions using `all`, `any`, `count`, `find`:

- `checkAllCustomersAreFrom` should return true if all customers are from a given city
- `hasCustomerFrom` should check if there is at least one customer from a given city
- `countCustomersFrom` should return the number of customers from a given city
- `findCustomerFrom` should return a customer who lives in a given city, or `null` if there is none

```kotlin
val numbers = listOf(-1, 0, 2)
val isZero: (Int) -> Boolean = { it == 0 }
numbers.any(isZero) == true
numbers.all(isZero) == false
numbers.count(isZero) == 1
numbers.find { it > 0 } == 2
```

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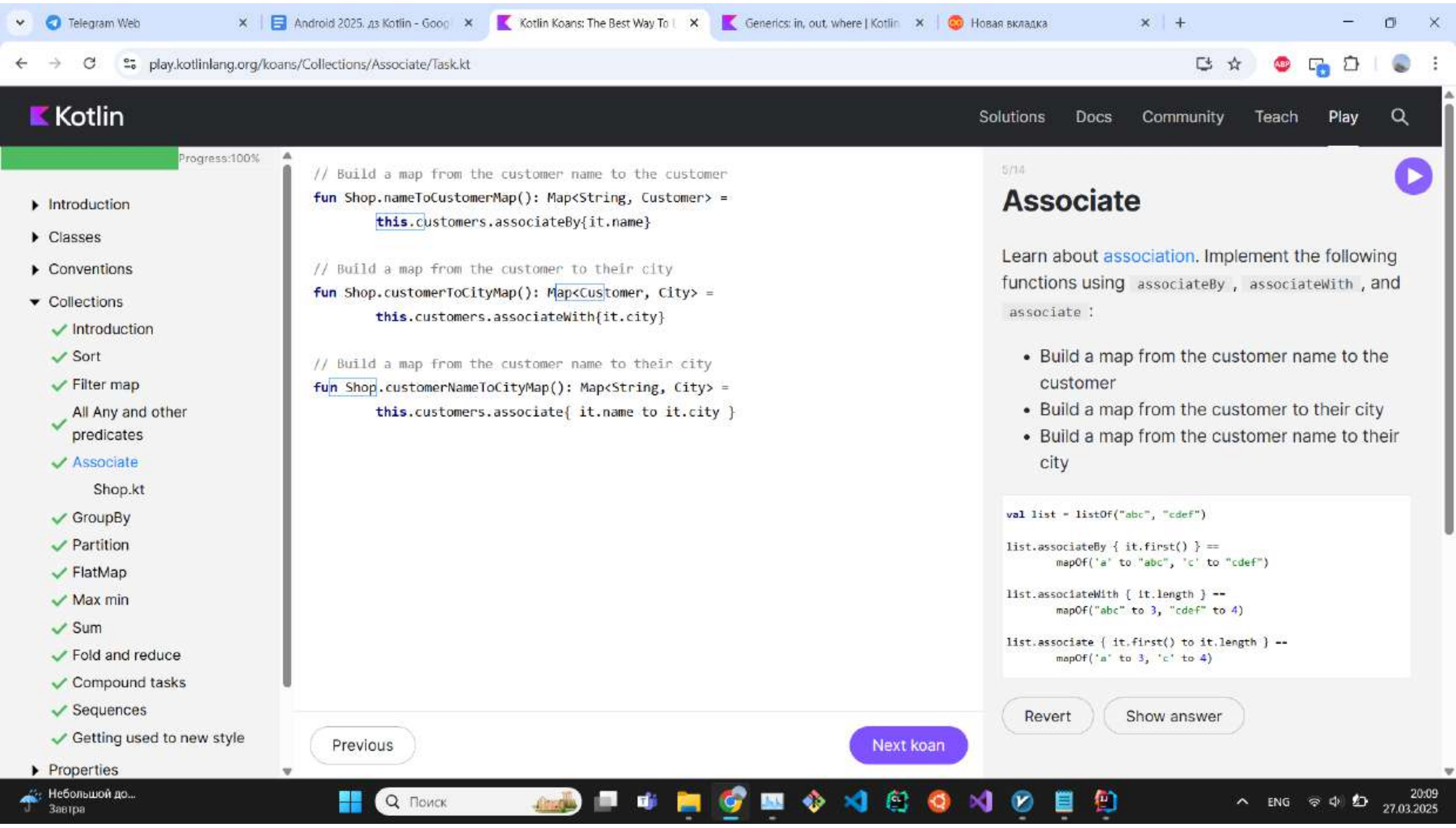

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```kotlin
// Build a map from the customer name to the customer
fun Shop.nameToCustomerMap(): Map<String, Customer> =
        this.customers.associateBy{it.name}

// Build a map from the customer to their city
fun Shop.customerToCityMap(): Map<Customer, City> =
        this.customers.associateWith{it.city}

// Build a map from the customer name to their city
fun Shop.customerNameToCityMap(): Map<String, City> =
        this.customers.associate{ it.name to it.city }
```

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Associate

Learn about association. Implement the following functions using `associateBy`, `associateWith`, and `associate`:

- Build a map from the customer name to the customer
- Build a map from the customer to their city
- Build a map from the customer name to their city

```kotlin
val list = listOf("abc", "cdef")

list.associateBy { it.first() } ==
        mapOf('a' to "abc", 'c' to "cdef")

list.associateWith { it.length } ==
        mapOf("abc" to 3, "cdef" to 4)

list.associate { it.first() to it.length } ==
        mapOf('a' to 3, 'c' to 4)
```

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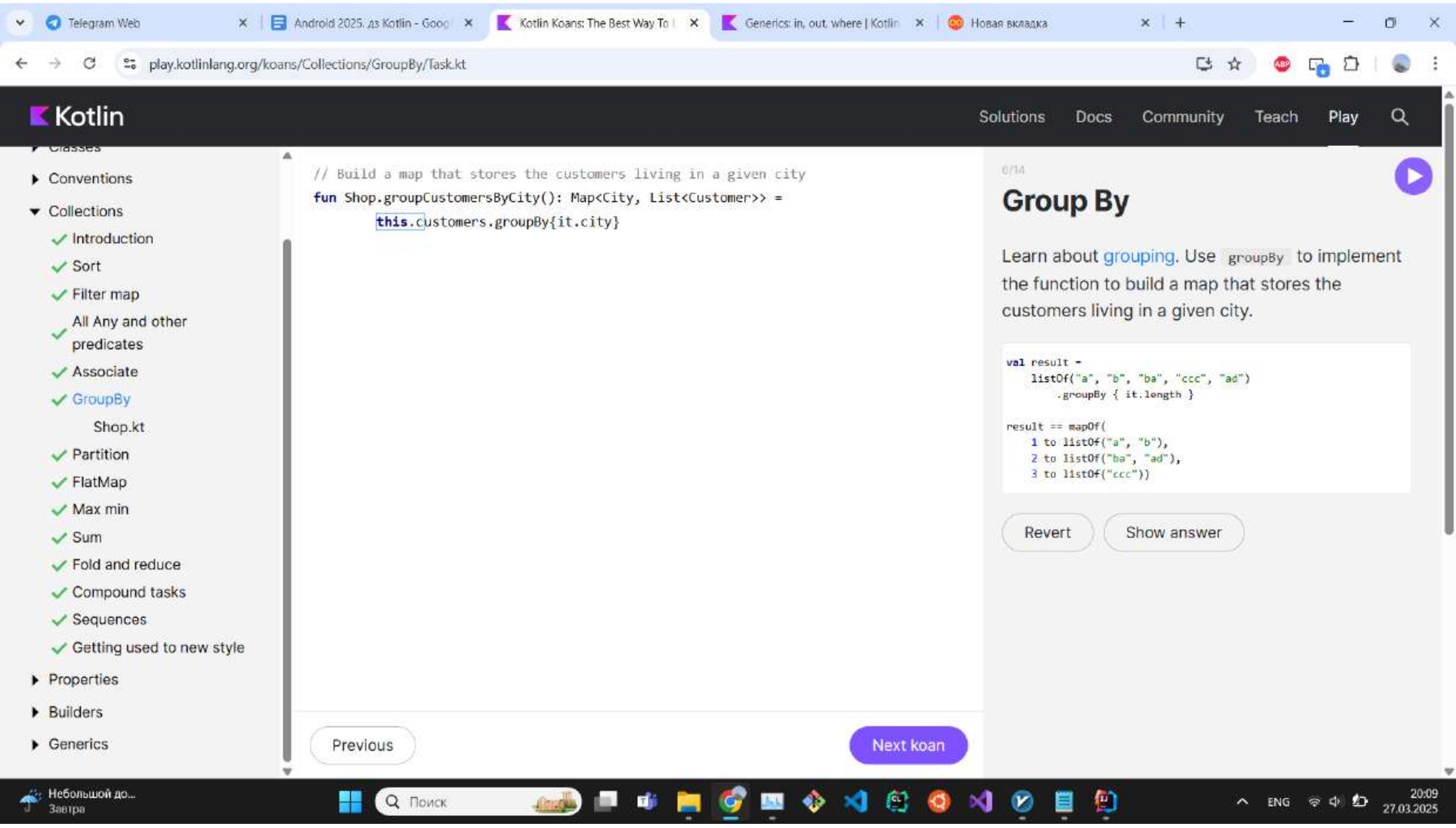
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// Build a map that stores the customers living in a given city
fun Shop.groupCustomersByCity(): Map<City, List<Customer>> =
this.customers.groupBy{it.city}
Group By
Learn about grouping. Use groupBy to implement the function to build a map that stores the customers living in a given city.
val result =
listOf("a", "b", "ba", "ccc", "ad")
.groupBy { it.length }
result == mapOf(
1 to listOf("a", "b"),
2 to listOf("ba", "ad"),
3 to listOf("ccc"))
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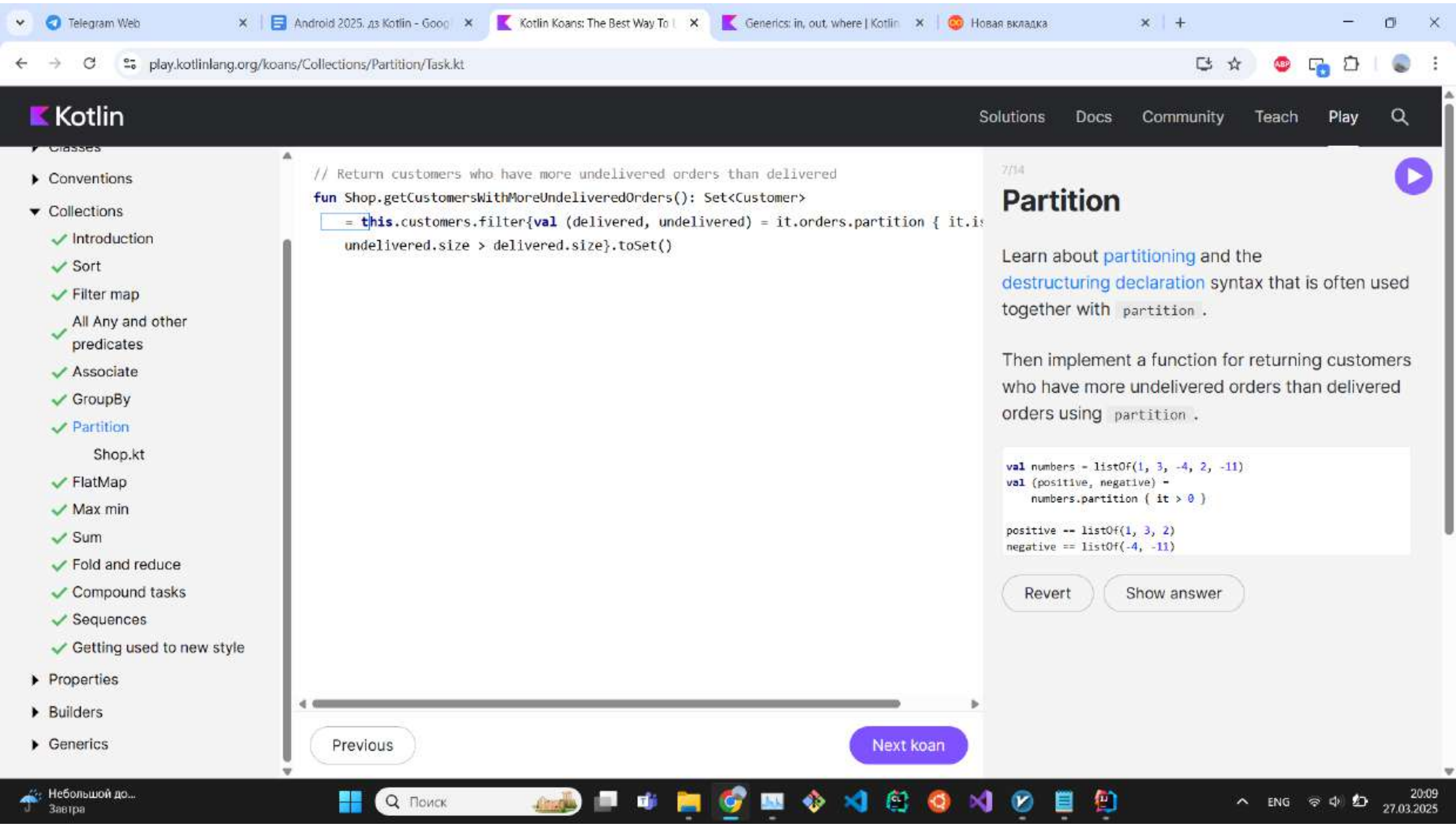

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// Return customers who have more undelivered orders than delivered
fun Shop.getCustomersWithMoreUndeliveredOrders(): Set<Customer>
= this.customers.filter{val (delivered, undelivered) = it.orders.partition { it.i
undelivered.size > delivered.size}.toSet()
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Partition
Learn about partitioning and the destructuring declaration syntax that is often used together with partition .
Then implement a function for returning customers who have more undelivered orders than delivered orders using partition .
val numbers = listOf(1, 3, -4, 2, -11)
val (positive, negative) =
numbers.partition { it > 0 }
positive == listOf(1, 3, 2)
negative == listOf(-4, -11)
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```kotlin
// Return all products the given customer has ordered
fun Customer.getOrderedProducts(): List<Product> =
        this.orders.flatMap{it.products}

// Return all products that were ordered by at least one customer
fun Shop.getOrderedProducts(): Set<Product> =
        this.customers.flatMap{it.orders.flatMap{it.products}}.toSet()
```

8/14

FlatMap

Learn about flattening and implement two functions using `flatMap`:

- The first should return all products the given customer has ordered
- The second should return all products that at least one customer ordered

```kotlin
val result = listOf("abc", "12")
    .flatMap { it.toList() }

result == listOf('a', 'b', 'c', '1', '2')
```

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```kotlin
// Return a customer who has placed the maximum amount of orders
fun Shop.getCustomerWithMaxOrders(): Customer? =
        this.customers.maxByOrNull{it.orders.size}

// Return the most expensive product that has been ordered by the given customer
fun getMostExpensiveProductBy(customer: Customer): Product? =
        customer.orders.flatMap{it.products}.maxByOrNull{it.price}
```

9/14

Max min

Learn about collection aggregate operations.

Implement two functions:

- The first should return the customer who has placed the most amount of orders in this shop
- The second should return the most expensive product that the given customer has ordered

The functions `maxOrNull`, `minOrNull`, `maxByOrNull`, and `minByOrNull` might be helpful.

```kotlin
listOf(1, 42, 4).maxOrNull() == 42
listOf("a", "ab").minByOrNull(String::length) == "a"
```

You can use callable references instead of lambdas. It can be especially helpful in call chains, where `it` occurs in different lambdas and has different types. Implement the `getMostExpensiveProductBy` function using callable references.

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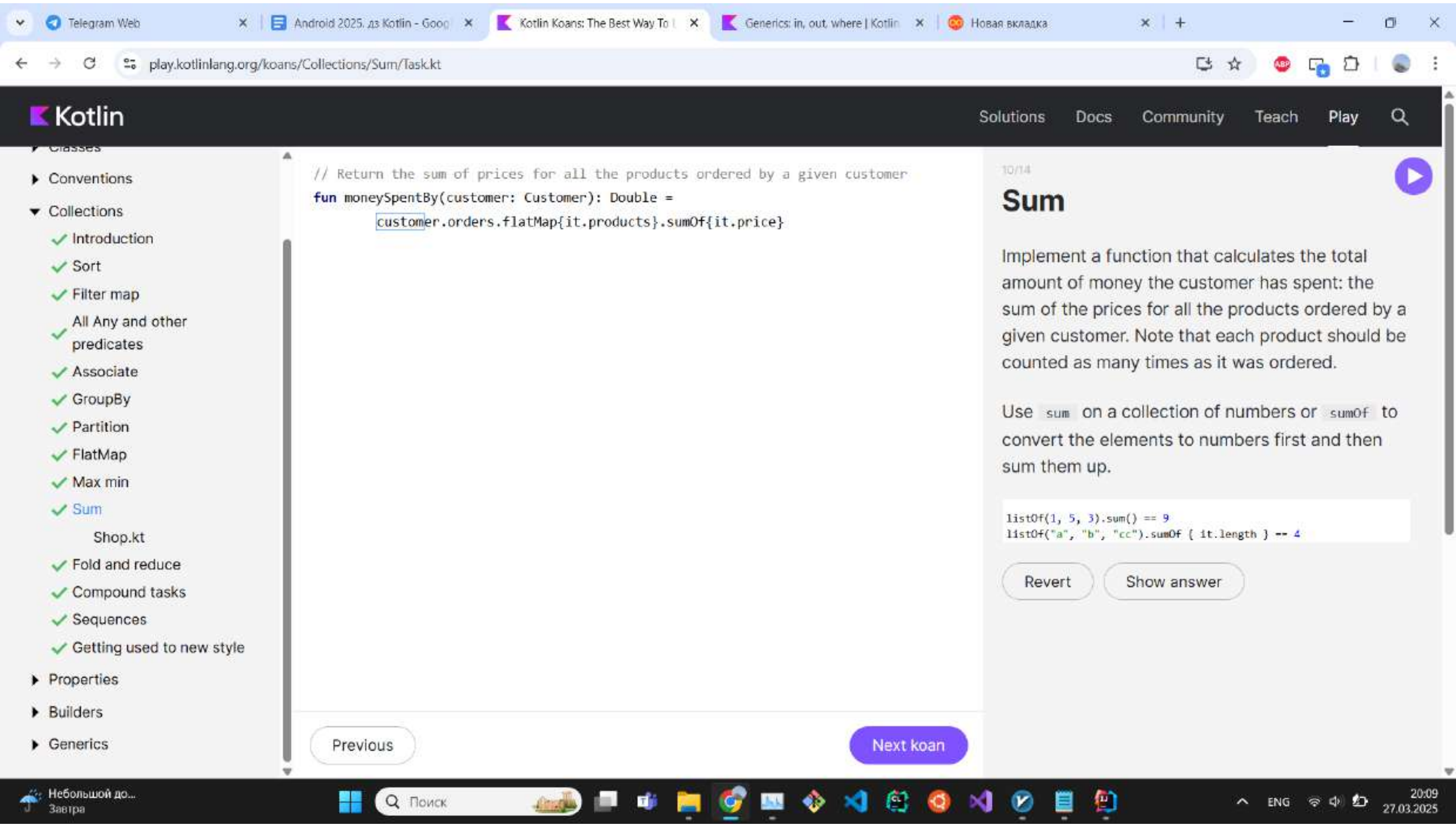
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// Return the sum of prices for all the products ordered by a given customer
fun moneySpentBy(customer: Customer): Double =
customer.orders.flatMap{it.products}.sumOf{it.price}
10/14
Sum
Implement a function that calculates the total amount of money the customer has spent: the sum of the prices for all the products ordered by a given customer. Note that each product should be counted as many times as it was ordered.
Use sum on a collection of numbers or sumOf to convert the elements to numbers first and then sum them up.
listOf(1, 5, 3).sum() == 9
listOf("a", "b", "cc").sumOf { it.length } == 4
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```kotlin
// Return the set of products that were ordered by all customers
fun Shop.getProductsOrderedByAll(): Set<Product> =
    this.customers.map{it.getOrderedProducts()}.reduce {res, nex -> res intersect nex}

fun Customer.getOrderedProducts(): Set<Product> =
    this.orders.flatMap {it.products}.toSet()
```

11/14

Fold and reduce

Learn about fold and reduce and set-specific operations and implement a function that returns the set of products that all the customers ordered using `reduce`.

You can use the `Customer.getOrderedProducts()` defined in the previous task (copy its implementation).

```kotlin
listOf(1, 2, 3, 4)
    .fold(1) { partProduct, element ->
        element * partProduct
    } == 24
```

You might also need the intersect function.

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```kotlin
// Find the most expensive product among all the delivered products
// ordered by the customer. Use `Order.isDelivered` flag.
fun findMostExpensiveProductBy(customer: Customer): Product? {
    return customer.orders.filter{it.isDelivered}.flatMap{it.products}.maxByOrNull{it.
}

// Count the amount of times a product was ordered.
// Note that a customer may order the same product several times.
fun Shop.getNumberOfTimesProductWasOrdered(product: Product): Int {
    return this.customers.flatMap{it.getOrderedProducts()}.count{it == product}
}

fun Customer.getOrderedProducts(): List<Product> =
        this.orders.flatMap {it.products}
```

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Compound tasks

Implement two functions:

- The first one should find the most expensive product among all the *delivered* products ordered by the customer. Use `Order.isDelivered` flag
- The second one should count the number of times a product was ordered. Note that a customer may order the same product several times

Use the functions from the Kotlin standard library that were previously discussed.

You can use the `Customer.getOrderedProducts()` function defined in the previous task (copy its implementation).

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```kotlin
// Find the most expensive product among all the delivered products
// ordered by the customer. Use `Order.isDelivered` flag.
fun findMostExpensiveProductBy(customer: Customer): Product? {
    return customer.orders.asSequence().filter{it.isDelivered}.flatMap{it.products}.ma
}

// Count the amount of times a product was ordered.
// Note that a customer may order the same product several times.
fun Shop.getNumberOfTimesProductWasOrdered(product: Product): Int {
    return this.customers.asSequence().flatMap{it.getOrderedProducts()}.count{it == pr
}

fun Customer.getOrderedProducts(): Sequence<Product> =
        this.orders.asSequence().flatMap {it.products}
```

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Sequences

Learn about sequences, they allow you to perform operations lazily rather than eagerly.

Copy the implementation from the previous task and modify it in a way that the operations on sequences are used.

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```kotlin
fun doSomethingWithCollection(collection: Collection<String>): Collection<String>? {

    val groupsByLength = collection.groupBy { s ->
        s.length
    }

    val maximumSizeOfGroup = groupsByLength.values.map { group ->
        group.size
    }.maxOrNull()

    return groupsByLength.values.firstOrNull {
        group -> group.size == maximumSizeOfGroup
    }
}
```

14/14

Getting used to the new style

We can rewrite and simplify the following code using lambdas and operations on collections. Fill in the gaps in `doSomethingWithCollection`, the simplified version of the `doSomethingWithCollectionOldStyle` function, so that its behavior stays the same and isn't modified in any way.

```kotlin
fun doSomethingWithCollectionOldStyle(
    collection: Collection<String>
): Collection<String>? {
    val groupsByLength = mutableMapOf<Int, MutableList<String>>()
    for (s in collection) {
        var strings: MutableList<String>? = groupsByLength[s.leng
        if (strings == null) {
            strings = mutableListOf()
            groupsByLength[s.length] = strings
        }
        strings.add(s)
    }

    var maximumSizeOfGroup = 0
    for (group in groupsByLength.values) {
        if (group.size > maximumSizeOfGroup) {
            maximumSizeOfGroup = group.size
        }
    }
```

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```
class PropertyExample() {
    var counter = 0
    var propertyWithCounter: Int? = null
        set (value) {
            field = value
            counter++
        }
}
```

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Properties

Learn about properties in Kotlin.

Add a custom setter to `PropertyExample.propertyWithCounter` so that the `counter` property is incremented every time the `propertyWithCounter` is assigned.

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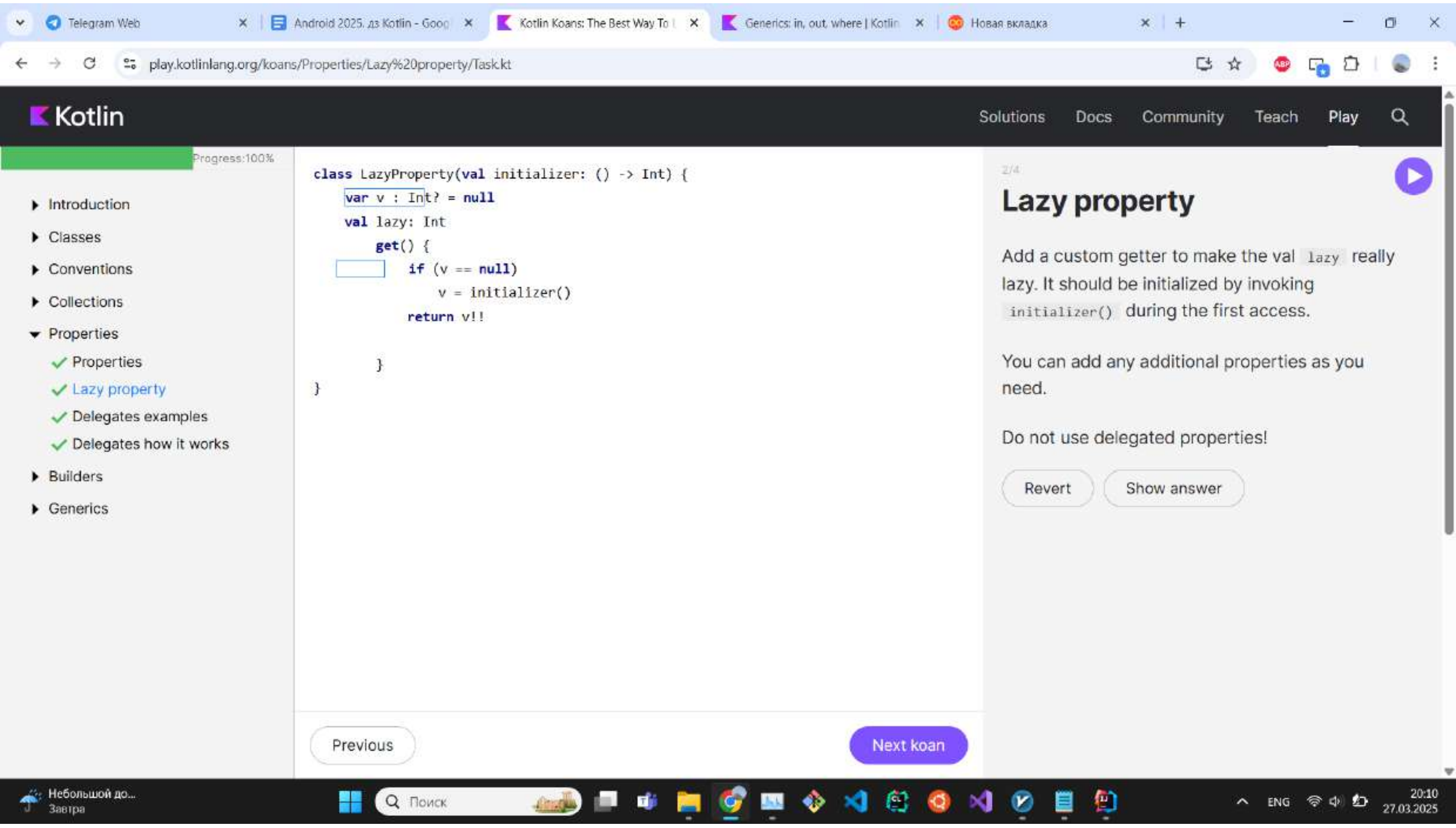

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class LazyProperty(val initializer: () -> Int) {
var v : Int? = null
val lazy: Int
get() {
if (v == null)
v = initializer()
return v!!
}
}
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Lazy property
Add a custom getter to make the val lazy really lazy. It should be initialized by invoking initializer() during the first access.
You can add any additional properties as you need.
Do not use delegated properties!
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```
import kotlin.reflect.KProperty
class LazyDelegate(val initializer: () -> Int) {
    var va : Int? = null
    operator fun getValue(v : Any?, p:KProperty<*>):Int {
        if (va == null)
            va = initializer()
        return va!!
    }
}

class LazyProperty(val initializer: () -> Int) {
    val lazyValue: Int by lazy(initializer)
}
```

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Delegates example

Learn about delegated properties and make the property lazy using delegates.

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```
import kotlin.properties.ReadWriteProperty
import kotlin.reflect.KProperty

class D {
    var date: MyDate by EffectiveDate()
}

class EffectiveDate<R> : ReadWriteProperty<R, MyDate> {

    var timeInMillis: Long? = null

    override fun getValue(thisRef: R, property: KProperty<*>): MyDate {
        return timeInMillis?.toDate() ?: MyDate(0,0,0)
    }

    override fun setValue(thisRef: R, property: KProperty<*>, value: MyDate) {
        timeInMillis = value.toMillis()
    }
}
```

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Delegates

You can declare your own delegates. Implement the methods of the class `EffectiveDate` so you can delegate to it. Store only the time in milliseconds in the `timeInMillis` property.

Use the extension functions `MyDate.toMillis()` and `Long.toDate()`, defined in `MyDate.kt`.

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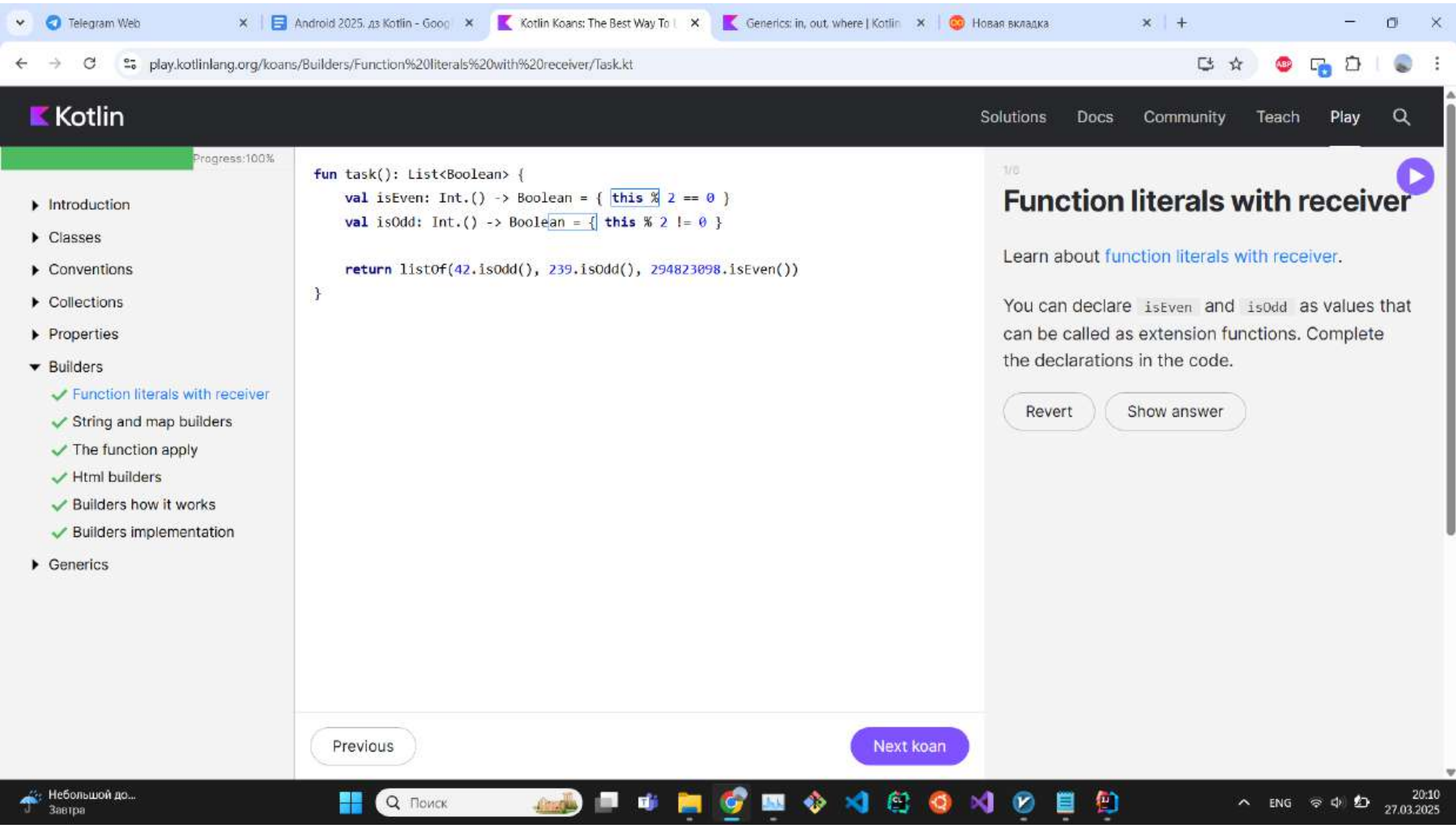
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fun task(): List<Boolean> {
val isEven: Int.() -> Boolean = { this % 2 == 0 }
val isOdd: Int.() -> Boolean = { this % 2 != 0 }
return listOf(42.isOdd(), 239.isOdd(), 294823098.isEven())
}
Function literals with receiver
Learn about function literals with receiver.
You can declare isEven and isOdd as values that can be called as extension functions. Complete the declarations in the code.
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```kotlin
import java.util.HashMap

fun buildMutableMap(b: HashMap<Int, String>.()->Unit): Map<Int, String> {
    var m :HashMap<Int, String> = HashMap()
    m.b()
    return m
}

fun usage(): Map<Int, String> {
    return buildMutableMap {
        put(0, "0")
        for (i in 1..10) {
            put(i, "$i")
        }
    }
}
```

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String and map builders

Function literals with receiver are very useful for creating builders, for example:

```kotlin
fun buildString(build: StringBuilder.() -> Unit): String {
    val stringBuilder = StringBuilder()
    stringBuilder.build()
    return stringBuilder.toString()
}

val s = buildString {
    this.append("Numbers: ")
    for (i in 1..3) {
        // 'this' can be omitted
        append(i)
    }
}

s == "Numbers: 123"
```

Implement the function `buildMutableMap` that takes a parameter (of extension function type), creates a new `HashMap`, builds it, and returns it as a result. Note that starting from 1.3.70, the standard library has a similiar `buildMap` function.

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```
fun <T> T.myApply(f: T.() -> Unit): T {
    this.f()
    return this
}

fun createString(): String {
    return StringBuilder().myApply {
        append("Numbers: ")
        for (i in 1..10) {
            append(i)
        }
    }.toString()
}

fun createMap(): Map<Int, String> {
    return hashMapOf<Int, String>().myApply {
        put(0, "0")
        for (i in 1..10) {
            put(i, "$i")
        }
    }
}
```

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The function apply

The previous examples can be rewritten using the library function `apply`. Write your implementation of this function named `myApply`.

Learn about the other scope functions and how to use them.

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```
fun renderProductTable(): String {
    return html {
        table {
            tr(color = getTitleColor()) {
                td {
                    text("Product")
                }
                td {
                    text("Price")
                }
                td {
                    text("Popularity")
                }
            }
            val products = getProducts()
            for ((index, product) in products.withIndex()) {
                tr {
                    td(color = getCellColor(index, 0)) {
                        text(product.description)
                    }
                    td(color = getCellColor(index, 1)) {
                        text(product.price)
                    }
                    td(color = getCellColor(index, 2)) {
                        text(product.popularity)
                    }
```

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HTML builder

1. Fill the table with proper values from the product list. The products are declared in `data.kt`.
2. Color the table like a chessboard. Use the `getTitleColor()` and `getCellColor()` functions. Pass a color as an argument to the functions `tr`, `td`.

Run the main function defined in the file `demo.kt` to see the rendered table.

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```kotlin
import Answer.*

enum class Answer { a, b, c }

val answers = mapOf<Int, Answer?>(
        1 to c, 2 to b, 3 to b, 4 to c
)
```

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Builders: how they work

Answer the questions below

1. In the Kotlin code

```kotlin
tr {
    td {
        text("Product")
    }
    td {
        text("Popularity")
    }
}
```

`td` **is:**

a. a special built-in syntactic construct

b. a function declaration

c. a function invocation

2. In the Kotlin code

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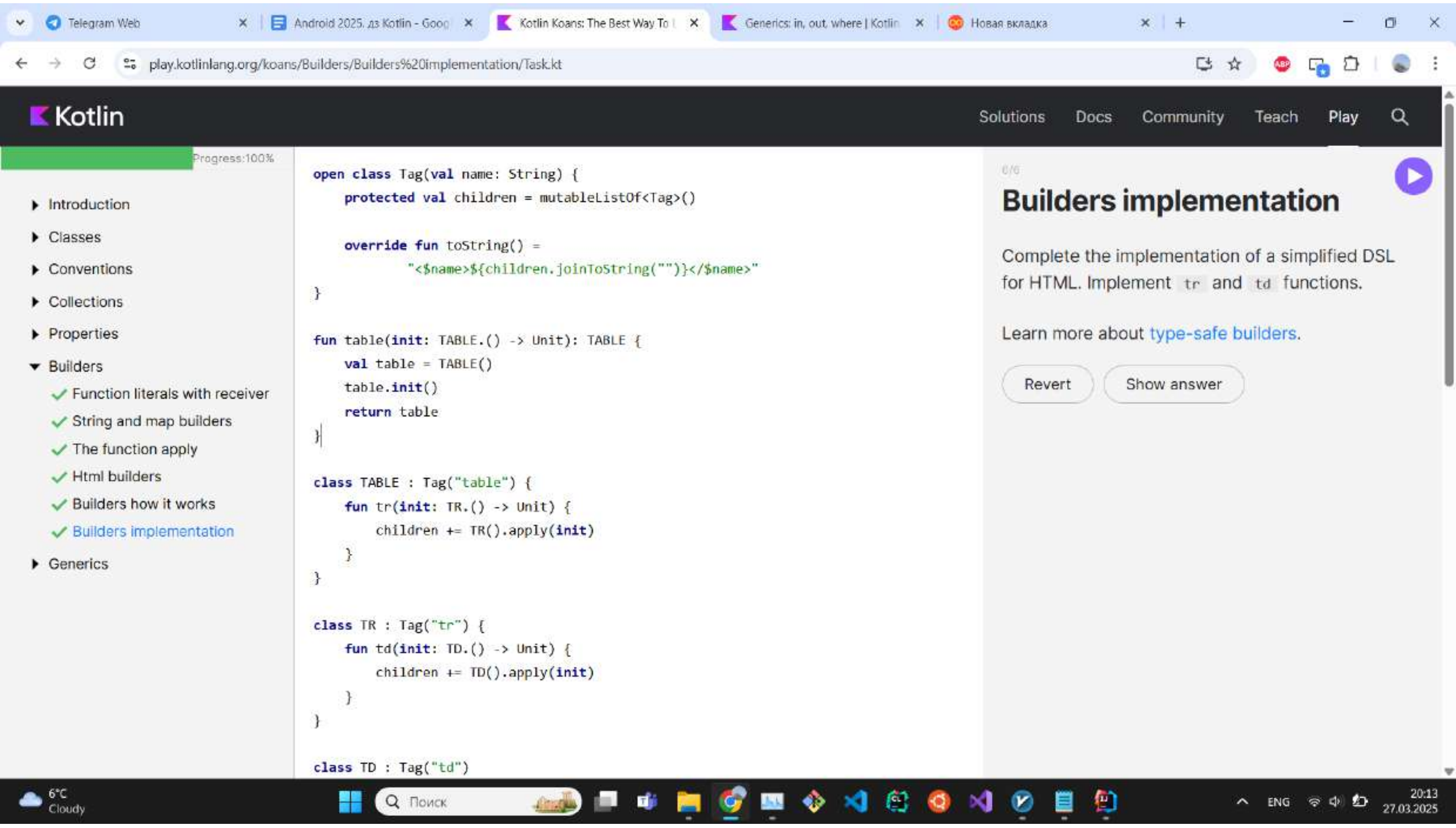
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open class Tag(val name: String) {
protected val children = mutableListOf<Tag>()
override fun toString() =
"<$name>${children.joinToString("")}</$name>"
}
fun table(init: TABLE.() -> Unit): TABLE {
val table = TABLE()
table.init()
return table
}
class TABLE : Tag("table") {
fun tr(init: TR.() -> Unit) {
children += TR().apply(init)
}
}
class TR : Tag("tr") {
fun td(init: TD.() -> Unit) {
children += TD().apply(init)
}
}
class TD : Tag("td")
Builders implementation
Complete the implementation of a simplified DSL for HTML. Implement tr and td functions.
Learn more about type-safe builders.
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```kotlin
import java.util.*

fun <T, C : MutableCollection<T>> Collection<T>.partitionTo(f: C, s: C, predicate: (T)
    for (el in this)
    {
        if (predicate(el))
            f.add(el)
        else
            s.add(el)
    }
    return f to s
}

fun partitionWordsAndLines() {
    val (words, lines) = listOf("a", "a b", "c", "d e")
            .partitionTo(ArrayList(), ArrayList()) { s -> !s.contains(" ") }
    check(words == listOf("a", "c"))
    check(lines == listOf("a b", "d e"))
}

fun partitionLettersAndOtherSymbols() {
    val (letters, other) = setOf('a', '%', 'r', '}')
            .partitionTo(HashSet(), HashSet()) { c -> c in 'a'..'z' || c in 'A'..'Z' }
    check(letters == setOf('a', 'r'))
    check(other == setOf('%', '}'))
}
```

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Generic functions

Learn about generic functions. Make the code compile by implementing a `partitionTo` function that splits a collection into two collections according to the predicate.

There is a `partition()` function in the standard library that always returns two newly created lists. Write a function that splits the collection into two collections given as arguments. The signature of the `toCollection()` function from the standard library might help you.

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